

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Alpha Home Tuition Platform

Connecting Verified Tutors with Parents & Students

Platform Version: 1.0.0 (Web & Mobile)

Client Organization: Alpha Home Tuition

Development Agency: Tech Maurya Team

Author: Product & Engineering Lead

Current Date: July 19, 2026

Document Status: Approved / Production-Ready

Document Control & Version History

This Product Requirements Document (PRD) outlines the comprehensive architectural, database, and functional specifications of the Alpha Home Tuition platform. It serves as the baseline for development, verification, deployment, and future scalability.

Version	Date	Author / Contributors	Description / Status
V1.0.0	July 19, 2026	Tech Maurya Team	Initial Draft. Comprehensive spec covering Supabase DB, PayPal Subscriptions, Custom Admin Matching engine, AI chatbot, and Capacitor integration. Status: Approved.

Table of Contents

1. Executive Summary & Project Background.....	3
2. Platform Target Audience & User Personas.....	3
3. System Architecture & Technical Stack.....	4
4. Detailed Functional Requirements.....	5
4.1 Marketing & Public-Facing Website.....	5
4.2 Multi-Persona Authentication & Onboarding.....	5
4.3 Six-Step Tutor Registration Wizard.....	6
4.4 Membership Subscriptions & Payments.....	6
4.5 Tutor Dashboard & Assignment Tracker.....	6
4.6 Super Admin Panel & Smart Matchmaker.....	7
4.7 Intelligent Hybrid AI Chatbot.....	7
5. Database Schema & Data Models.....	8
6. Non-Functional & Native Compilation Requirements.....	9
7. Future Roadmap & Product Scalability.....	10

1. Executive Summary & Project Background

Alpha Home Tuition is a premium home tuition mediator platform designed to connect students and parents with background-checked, highly-qualified tutors in their local vicinity or online. The home tuition industry is highly fragmented, suffering from lack of transparency, security, and high mediator commissions. Alpha Home Tuition addresses these gaps by implementing a reliable database-driven matchmaker model, coupled with a robust tutor-side premium subscription.

1.1 The Core Problem & Challenges

Safety & Verification: Parents struggle to find trusted tutors without security concerns or credential fraud.

Mediator Fees: Traditional agencies charge up to 50% of the tutor's first month salary, demotivating tutors.

Inefficient Matching: Finding a tutor with specific combinations of subjects, class boards, locations, and budget range is tedious and manual.

1.2 The Platform Solution

The Alpha Home Tuition digital platform addresses these issues directly. Tutors register with academic records, target locations, expected salary, and upload verification files. Parents and students request trial demo classes with exact constraints. Super Admins quickly screen applications, match, and assign verified tutors. The platform generates revenue through tutor membership subscriptions (Silver/Platinum).

2. Platform Target Audience & User Personas

The platform supports three distinct types of users, each with specialized interfaces, authentication, and data access levels:

Students & Parents: Primary consumers of tuition. They require an easy-to-use search and request interface. They submit demo requests specifying class level, syllabus boards, home address, budget, and subject clusters. They do not pay registration fees; they only pay the agreed salary directly to tutors.

Tutors: Experienced educators, graduates, or students seeking part-time/full-time tutoring roles. They register through a detailed profile wizard, purchase a subscription (Silver or Platinum) via a PayPal integrated checkout, and use their dashboard to track assigned trial jobs and earnings.

Super Administrators: Internal operations team at Alpha Home Tuition. They log in via a secure admin portal to review tutor dossiers, inspect physical documents, match and assign tutors to incoming student demo requests, and monitor overall payouts, bills, and communications.

3. System Architecture & Technical Stack

The platform is built on a serverless, decoupling-first architecture. A React single-page frontend communicates securely with Supabase backend services, utilizing client-side route guards, database triggers, and cloud Edge Functions. The web build is natively packaged into an Android application.

3.1 Technical Stack Breakdown

Core SPA Framework: React 19 + Vite 8. React 19 provides highly responsive rendering, while Vite ensures instant development hot-reloads and ultra-optimized production bundles.

Design & Interactions: Tailwind CSS v4 for utility-first layout construction. Framer Motion handles beautiful entrance/scroll transitions and accordion layouts. Lucide React provides modern UI icons.

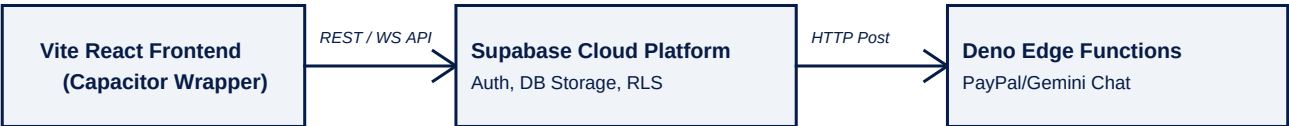
Backend (Supabase): PostgreSQL database for reliable transactional storage. Supabase Auth manages user credentials and secure session tokens. Supabase Storage hosts documents and avatars.

Serverless Compute: Supabase Edge Functions written in Deno (TypeScript). These handle the AI chatbot engine ('chat') and process billing transactions asynchronously ('paypal-webhook').

Mobile Wrapper: Capacitor JS v8. Integrates Android plugins, maps the web build into native directories, and allows compiling into a production APK/AAB bundle.

Gateway Integrations: PayPal SDK for processing subscription payments on the client. EmailJS handles direct email notifications for contact forms.

3.2 Architectural Flow Concept



Row Level Security (RLS) policies restrict database read/write queries based on the user's authenticated role (e.g. parent, tutor, admin). This prevents unauthorized database modifications.

4. Detailed Functional Requirements

4.1 Marketing & Public-Facing Website

The visitor website focuses on user acquisition (converting parents to submit demo requests, and tutors to register). The requirements include:

Sticky Navigation: A floating header responsive to mobile hamburger interactions. Includes links to Home, About, Services, Contact, and Join Us.

Hero Stat Banner: A dark navy hero with smooth Framer Motion transitions. Displays dynamic counters (e.g., '500+ Verified Tutors', '1200+ Students Helped', '98% Success Rate').

Interactive Contact Form: Collects name, email, phone number, and message. Directly sends an email copy via EmailJS client SDK and saves the record in 'contact_messages' database table.

Floating Actions: A floating WhatsApp button linking to the support team (Phone: +91 7295948480) and a smooth scroll-to-top button.

4.2 Multi-Persona Authentication & Onboarding

Authentication is decoupled by roles. The registration flows require distinct validation rules:

Parent & Student Access: Parents register with Email, Password, Name, and Contact Number. They gain access to '/student' route, showing a personal dashboard to manage requested demo classes.

Tutor Profile Creation: Tutors register their initial account via '/tutor/register'. They must complete the 6-Step onboarding profile wizard before accessing the dashboard.

Admin Entry: Access to '/admin/login' is protected. The login credentials verify if the profile role equals 'admin', granting clearance to '/admin/dashboard'.

4.3 Six-Step Tutor Registration Wizard

To ensure high standards of academic quality, tutors must complete a detailed multi-step profile application. Each step is fully validated on the client side:

Step 1: Personal Details: Full name, validated 10-digit mobile number, email address, physical address, city, and gender selection.

Step 2: Education Profile: 10th board marks, 12th board marks, degree classification (e.g., B.Tech, B.Sc, M.A.), college name, and specialized subjects studied.

Step 3: Experience & Preferences: Years of active teaching experience, selected classes preferred (Primary, Middle, High School), subjects comfortable teaching, and expected monthly salary range.

Step 4: Document Dossier: Upload fields for passport photo, Aadhar Card (identity proof), and academic marksheets/degree certificates. Documents are stored inside Supabase Storage buckets.

Step 5: Membership Selection: Selection of Silver (Rs. 500/year) or Platinum (Rs. 1,000/year) plan. Direct call-to-action to process digital registration fee.

Step 6: Compliance Declaration: Explicit checkbox to accept terms of service, refund policy, and code of conduct. Submission inserts profile data into the database in 'Pending' status.

4.4 Membership Subscriptions & Payments Flow

Tutors pay a one-time annual registration fee. The features include:

Pricing & Tiers: Silver Plan (Rs. 500, expected salary Rs. 1500-2500 per tuition). Platinum Plan (Rs. 1000, expected salary Rs. 3000+, certificate of appreciation, priority support, popular badge).

PayPal Checkout: Integrated via '@paypal/react-paypal-js'. Tutors execute the payment directly. Upon client approval, the payment transaction is completed.

Supabase PayPal Webhook: Deno Edge Function 'paypal-webhook' receives callbacks, verifies transaction signature, sets membership to active, updates expiry date (current + 365 days), and creates a row in 'tutor_memberships' table.

4.5 Tutor Dashboard & Assignment Tracker

Once verified by the administrator, tutors gain access to their workspace '/dashboard':

Status Overview: Shows tutor profile status (Pending review, Verified, Rejected).

Active Assignments: Lists matching students and demo class schedules assigned by the administrator.

Earnings Ledger: Shows total earnings generated per assignment, track pending/cleared payouts.

4.6 Super Admin Panel & Smart Matchmaker

The control center of the platform runs on '/admin/dashboard'. It empowers administrative staff to manage core transactions:

Dashboard Overview Stats: Renders total tutors, pending registrations, pending student demo requests, and active matches. Features visual summaries.

Tutor Directory & Approval: Displays searchable database of tutors. Admin can open details modal, inspect uploaded academic credentials, and toggle status to 'Verified' or 'Rejected'.

Demo Requests Manager: Lists parents who requested home/online classes. Statuses: Pending, Contacted, Tutor Assigned, Completed. Staff can update details on the fly.

Smart Tutor Matchmaker: When an admin clicks 'Assign Tutor' on a demo request, the matching engine searches the 'tutors' table for verified teachers matching: 1. Target City & Location, 2. Desired Class Level, 3. Subject Specialization, and 4. Gender preferences. Lists potential fits. One-click assignment creates a record in 'tutor_assignments' and notifies the tutor.

4.7 Intelligent Hybrid AI Chatbot

The landing pages feature an active floating chatbot. It maximizes conversion rates by resolving parent queries immediately:

Keyword FAQ Engine: Tokenizes query, strips stop-words (e.g., 'the', 'how', 'is'), and runs a weighted matching score against static local FAQ data. High-ranking items return instant answers without server latency.

Gemini Edge Fallback: If score is low, queries are forwarded to Deno Edge Function 'chat'. It sends the prompt to Gemini LLM with context on Alpha Home Tuition (fees, safety, locations) and returns natural responses.

5. Database Schema & Data Models

The PostgreSQL database design enforces relational integrity, indexing, and cascade delete rules. Below are the structures of the key tables:

5.1 Table: profiles

Column Name	Data Type	Constraints & Description
id	uuid	Primary Key (references auth.users.id)
role	text	Role classification: 'tutor' 'admin' 'student'
full_name	text	Required field. Full name of the user
email	text	Required field. Email address for communication
phone	text	Contact mobile number
created_at	timestamp	Default: timezone('utc'::text, now())

5.2 Table: tutors

Column Name	Data Type	Constraints & Description
id	uuid	Primary Key (references profiles.id, CASCADE)
full_name	text	Required. Name of the tutor
email	text	Required. Unique contact email
phone	text	Required. 10-digit primary mobile number
city	text	Required. Primary city of tutor
qualification	text	Highest qualification (B.Tech, M.A. etc)
experience	integer	Years of tutoring experience
status	text	Verification state: 'Pending' 'Verified' 'Rejected'
membership	text	Selected tier: 'Silver' 'Platinum' 'None'
expected_salary	numeric	Expected minimum salary per month/class
created_at	timestamp	Creation timestamp

5.3 Table: demo_requests

Column Name	Data Type	Constraints & Description
id	uuid	Primary Key (Generated UUID)

Column Name	Data Type	Constraints & Description
student_name	text	Required. Student/parent name
class	text	Target class grade (e.g. 'Class 10')
subjects	text	Comma-separated subject strings
mode	text	Session mode: 'Home' 'Online' 'Offline'
location	text	Detailed sector address
contact_number	text	Validated mobile number
status	text	State: 'Pending' 'Contacted' 'Tutor Assigned' 'Completed'
created_at	timestamp	Submission timestamp

5.4 Table: tutor_assignments

Column Name	Data Type	Constraints & Description
id	uuid	Primary Key
demo_request_id	uuid	Foreign Key (references demo_requests.id, CASCADE)
tutor_id	uuid	Foreign Key (references tutors.id, CASCADE)
status	text	Assignment state: 'Assigned' 'Active' 'Completed'
notes	text	Matching remarks or guidelines by admin
assigned_by	uuid	Admin profiles.id reference
created_at	timestamp	Assignment date

5.5 Table: tutor_memberships

Column Name	Data Type	Constraints & Description
id	uuid	Primary Key
tutor_id	uuid	Foreign Key (references tutors.id, CASCADE)
plan	text	Subscription Tier: 'Silver' 'Platinum'
payment_id	text	PayPal transaction order reference ID
amount	numeric	Subscription fee paid
payment_status	text	State: 'COMPLETED' 'PENDING' 'FAILED'
expires_at	timestamp	Membership validation expiry timestamp
created_at	timestamp	Transaction timestamp

6. Non-Functional & Mobile Integration Requirements

6.1 Security & Access Control

Database RLS: Postgres Row Level Security (RLS) is active. Tutors can update only their 'tutors' record. Admins bypass policy for system operations.

Route Guards: The React frontend implements '<ProtectedRoute requiredRole="...">' wraps. Unauthorized URL direct inputs redirect to '/login'.

Credential Safety: Supabase Auth encrypts passwords. Token verification follows JWT standards.

6.2 Performance & Responsiveness

Page Speeds: Webpack-level lazy loading partitions large pages. First contentful paint (FCP) must remain under 1.5 seconds on mobile connections.

Media Control: Resumes and credentials are saved in raw PDF/JPEG formats. Pre-upload image compression filters down avatar sizes to < 500KB.

6.3 Capacitor Android Native Porting

The project includes Capacitor config files ('capacitor.config.ts'). The production build process requires compiling Vite assets, copying them to the Android source root, and triggering Android Studio:

Build Assets: npm run build (Compiles code into the static '/dist' folder).

Sync Native App: npx cap sync (Synchronizes Vite assets with the '/android' app structure).

Open Android Studio: npx cap open android (Opens IDE to generate the final debug or release APK).

7. Future Roadmap & Product Scalability

To scale the platform beyond a basic mediator system, the following features are planned for subsequent major release cycles (V2.0.0):

Milestone / Feature	Target Timeline	Technical Requirements	Impact Level
Native Push Notifications	Q4 2026	Firebase Cloud Messaging (FCM) integration with Capacitor plugins.	High
Automated Geo-Matching	Q1 2027	PostGIS queries in PostgreSQL to match tutor location with student coordinates.	Critical
In-App Chat Messenger	Q2 2027	Websockets / Supabase Realtime channels connecting parents and tutors.	Medium
Virtual Classrooms	Q4 2027	WebRTC integration or Zoom APIs embedded in student profiles.	High

8. Approval & Sign-Off

This document represents the official specifications of the Alpha Home Tuition Platform (V1.0.0). By signing below, stakeholders agree that the current implementation meets the requirements.

Alpha Home Tuition Representative

Tech Maurya Development Lead

Signature & Date

Signature & Date